

|                         |                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Ex. No. 1</b>        | WORKING WITH WIDGETS AND LAYOUTS                                                                              |
| <b>Date of Exercise</b> | 11.08.2026                                                                                                    |
| <b>GitHub Host Link</b> | <a href="https://github.com/Urk23cs1226/exp1-flutter.git">https://github.com/Urk23cs1226/exp1-flutter.git</a> |

### Aim

To develop a Flutter application using basic widgets and layout widgets such as Container, Row, Column, Expanded, Card, and List View to create a simple student profile/dashboard interface.

### Description

Flutter provides various widgets to design user interfaces. In this experiment, basic widgets such as Text, Icon, Container, Row, Column, Card, List Tile, and List View are used to create a student profile application. The application demonstrates how widgets can be arranged using different layout structures.

### Procedure:

1. Open **VS Code** and launch the Terminal.
2. Check whether Flutter is installed correctly using:

```
flutter doctor
```

3. Check the available devices using:

```
flutter devices
```

4. Create a new Flutter project using:

```
flutter create widget_layout
```

5. Move into the project directory:

```
cd widget_layout
```

6. Open the project in VS Code:

```
code .
```

7. Run the default Flutter application:

```
flutter run
```

8. Open the **lib/main.dart** file and delete the existing code.
9. Import the Flutter Material package and create the basic application using `MaterialApp`, `Scaffold`, `AppBar`, and `Text`.
10. Replace the body with a **Column** widget and add student information such as student name, department, and laboratory name.
11. Add a **Container** widget with padding and margin to display the student profile.
12. Add a **Row** widget with different icons such as Home, Person, and Settings using `MainAxisAlignment.space Evenly`.
13. Add **Card** widgets with `ListTile` to display information such as Flutter and Mobile Application Development.
14. Add a **ListView** containing multiple `ListTile` items such as:
  - Flutter
  - Cloud Computing
  - Programming
15. Save the program and run the application using:

```
flutter run
```

16. Observe the student profile/dashboard interface displayed on the selected device.

## **Program**

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Profile',
      theme: ThemeData(
```

```
colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),

useMaterial3: true,

),

home: const StudentProfile(),

);

}

}

class StudentProfile extends StatelessWidget {

  const StudentProfile({super.key});

  @override

  Widget build(BuildContext context) {

    return Scaffold(

      appBar: AppBar(

        title: const Text('Student Profile'),

        backgroundColor: Colors.blue,

        foregroundColor: Colors.white,

      ),

      body: ListView(

        padding: const EdgeInsets.all(16),

        children: [

          Container(

            padding: const EdgeInsets.all(20),

            margin: const EdgeInsets.only(bottom: 15),

            decoration: BoxDecoration(

              color: Colors.blue.shade50,

              borderRadius: BorderRadius.circular(12),

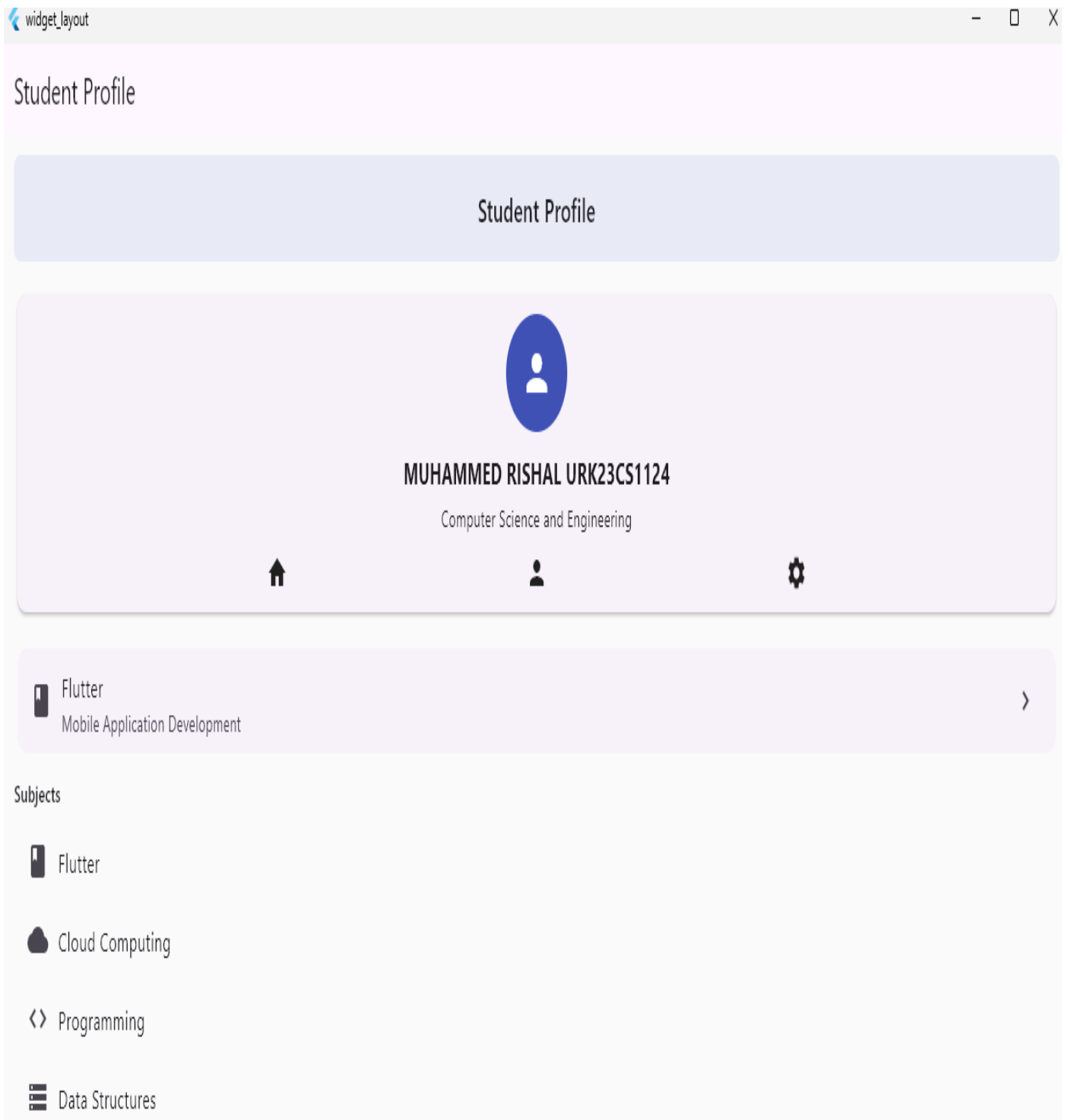
            ),
```

```
child: const Column(  
  children: [  
    Icon(  
      Icons.person,  
      size: 70,  
      color: Colors.blue,  
    ),  
    SizedBox(height: 10),  
    Text(  
      'Student Profile',  
      style: TextStyle(  
        fontSize: 24,  
        fontWeight: FontWeight.bold,  
      ),  
    ),  
    SizedBox(height: 5),  
    Text(  
      'URK23CS1226',  
      style: TextStyle(fontSize: 18),  
    ),  
    Text(  
      'Computer Science and Engineering',  
      textAlign: TextAlign.center,  
    ),  
  ],  
)
```

```
),  
const Text(  
  'Quick Access',  
  style: TextStyle(  
    fontSize: 20,  
    fontWeight: FontWeight.bold,  
  ),  
,  
,  
const SizedBox(height: 10),  
Row(  
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,  
  children: const [  
    Icon(Icons.home, size: 35, color: Colors.blue),  
    Icon(Icons.person, size: 35, color: Colors.green),  
    Icon(Icons.settings, size: 35, color: Colors.orange),  
  ],  
,  
const SizedBox(height: 20),  
const Card(  
  child: ListTile(  
    leading: Icon(Icons.book, color: Colors.blue),  
    title: Text('Flutter'),  
    subtitle: Text(  
      'Mobile Application Development',  
    ),  
  ),  
,  
,  
,
```

```
const Card(  
  child: ListTile(  
    leading: Icon(Icons.cloud, color: Colors.blue),  
    title: Text('Cloud Computing'),  
  ),  
,  
const Card(  
  child: ListTile(  
    leading: Icon(Icons.code, color: Colors.blue),  
    title: Text('Programming'),  
  ),  
,  
],  
,  
);  
}
```

## Output



The screenshot shows an IDE with the following components:

- EXPLORER:** A file tree on the left showing the project structure. The 'lib' directory is expanded, showing 'main.dart' as the active file.
- main.dart:** The main code file is open, showing a Flutter widget class named 'MyApp'. The code defines a 'build' method that returns a 'Row' widget. Inside the 'Row', there is a 'Column' widget containing a 'CircleAvatar' (with a person icon), a 'Text' widget (displaying 'MUHAMMED RISHAL URK23CS1124'), and another 'Text' widget (displaying 'Computer Science and Engineering').
- OUTPUT:** The bottom panel shows the output of the application. It includes a log entry for 'neVersion' and a 'CategoryInfo' entry with an 'InvalidOperationException' message, indicating a runtime error.

## Result

Thus, a Student Profile/Dashboard application was successfully developed using Flutter with Container, Row, Column, Card, List View, and other basic widgets. The application was executed successfully and the expected user interface was displayed.

